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Enhanced photovoltaic performance of silicon-based solar cell through optimization of Ga-doped ZnO layer

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Abstract

In the present study, the impact of deposition pressure and substrate temperature of Ga-doped Zinc Oxide (GZO) thin film and the photovoltaic performance of this structure as a transparent conductive oxide (TCE) layer in silicon-based solar cell were investigated. Implementing a single target of GZO, the structural, optical, and electrical properties of 350 nm thick GZO thin films with various deposition pressure (5 mTorr, 10 mTorr, 15 mTorr and 20 mTorr) at room temperature (RT) and substrate temperature (RT, 150 °C, 200 °C, 250 °C) at 15 mTorr deposition pressure were fabricated using RF magnetron sputtering technique. The aim here was to find out the GZO films with the optimum pressure and substrate temperature to incorporate them into solar cell as a TCE layer. The X-ray diffraction (XRD) and atomic force microscopy (AFM) techniques were used to determine the structural properties of all samples. The optical transmission measurements were performed using spectroscopic Ellipsometer and the band gap values were calculated by Tauc plot using optical transmission data. In addition, the electrical characterization of the GZO samples were analyzed by the Van der Pauw method and Hall measurements. Finally, the most promising GZO thin film was determined based on the structural and optoelectrical characterization. The findings indicated that the XRD pattern of all the prepared films was dominated by (002) preferential orientation irrespective of the deposition pressure and substrate temperature. The AFM measurements showed that all the samples had a dense surface morphology regardless of the deposition pressures, but the surface morphology of the samples was clearly changed upon increasing substrate temperatures. The transmission values of the film did not significantly alter (~82%) when the deposition pressures except for the substrate temperature of 200 °C (86%) were changed. The band gap values were calculated between 3.30 eV and 3.36 eV, which can be associated with enhancement of crystalline quality of the films. The lowest resistivity and the highest carrier concentration values belonged to the film fabricated at 15 mTorr@200 °C by 2.0×10^{-3} Ω.cm and 1.6×10^{20} cm^{-3}, respectively. Both increasing the deposition pressure (up to 15 mTorr) and substrate temperature (up to 200 °C) contributes to improving the crystallite size, widening the optical band gap, lowering the resistivity, and increasing the carrier concentration. In order to evaluate and compare the effect of both deposition pressure and substrate temperature, Silicon-based solar cells were fabricated using the most promising layers (15 mTorr@RT, 15 mTorr@200 °C). The cell performance with the GZO thin film as a TCE layer showed that varying both the pressure and substrate temperature of the GZO film contributed to enhancing the solar cell parameters. Thus, the conversion efficiency increased from 9.24% to 12.6% with the sequential optimization of pressure and temperature. It can be concluded that the pressure applied during the deposition and substrate temperature had a significant impact on the properties of GZO thin films and its photovoltaic performance of solar cell used as TCE layer.

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1. Introduction

Zinc Oxide (ZnO) and Tin Oxide (SnO_2) based transparent conducting electrode (TCE) thin films have widespread application areas since they consist of environmental-friendly, earth abundant, inexpensive, and chemically-stable raw materials [1]. Some of the common application areas include laser diodes [2], solar cells [3], sensors [4], transparent thin film transistors [5] and optical detectors [6]. However, the electrical properties of pristine ZnO and SnO_2 may limit the use of this material. Therefore, there have been studies on ZnO and SnO_2 doping with using aluminum (AZO) [7], molybdenum (MZO) [8], gallium (GZO) [9], fluorine (FTO) [10], Indium (ITO), etc elements in order to enhance its electrical and optical properties. In addition to ZnO-based TCE films, indium tin oxide (ITO) film is also a commonly employed material as a TCE layer because of its low electrical resistivity ($< 4 \times 10^{-4}$ Ω.cm) and high optical transmission (>80%) [11]. Amongst doped-ZnO materials, AZO can compete with the ITO material thanks to its outstanding electrical and optical features [9]. This thin film compound has 3.3–3.6 eV optical band gap and n-type electrical conductivity, which makes this material suitable as a TCE layer for solar cell applications [12]. However, there are some problems associated with both ITO and AZO compounds. ITO is a high cost material along with its brittle nature that may limit flexible applications of such compound [13]. Regarding AZO thin film, it also has some drawbacks such as strong photon absorption in the infrared (IR) region that may give rise to the reduction in the photo-conversion efficiency of the solar cells [14].

To overcome the aforementioned issues regarding ITO and AZO films, gallium doped zinc oxide (GZO) has recently been researched intensely as a TCE layer in the literature. Besides being an eco-friendly and low cost raw material, the GZO film has low resistivity, high transparency, and good stability [15]. Moreover, the ionic radii of Ga is 0.62 Å and its covalent radii is 1.26 Å, which are very close to the radii of Zn atom [16]. Hence, Ga^{3+} could be replaced with Zn^{2+} without effecting the ZnO host lattice. Furthermore, the length of covalent bond in Ga–O (1.92 Å) and Zn–O is competitive [17]. While the Al has a high activity that causes oxidation during the deposition process, Ga exhibits a relatively more resistant nature to oxidation [18]. The mentioned properties make Ga one of the most promising dopant in ZnO compound. When Ga is used as a doping material in ZnO, it contributes to shifting the band gap (3.3 eV–3.63 eV) and increasing the carrier concentration over 10^{20} cm^{-3} [19, 20].

To fabricate GZO thin films, several growth techniques such as electron beam evaporation [21], sol–gel [22], spray pyrolysis [23], pulsed laser deposition [24], atomic layer deposition [25], and radio frequency (RF) magnetron sputtering [17] can be implemented. The magnetron sputtering deposition approach is a preferable deposition technique for the production of ZnO-based TCE films with high quality since it provides homogenous and pure film together with its applicability to obtain films on large area, strong adhesion to the substrate, and precise controllable film thickness [26].

Lin *et al* prepared AZO, GZO, AGZO TCE layers with pulsed laser deposition method (PLD). They applied and compared AGZO, AZO and GZO thin films as a TCE layer for Si based solar cells. They reported that the best cell efficiency belonged to the AGZO with 7.34% cell efficiency. When they employed GZO and AZO separately, the cell efficiencies were found to be 6.45% and 5.48%, respectively (Lin *et al* 2012). In addition, Yan *et al* fabricated GZO thin films by reactive plasma deposition for Si hetero-junction (SHJ) solar cells. After the characterization steps, they put forward that GZO layer is more suitable for Si based SHJ solar cells as a TCE layer with 23.65% cell efficiency (Yan *et al* 2023). As seen, the efficiency of the cells with GZO films as TCE can also be affected by the film deposition parameters [27, 28].

The effect of thickness [29], dopant concentration [30], post-annealing [31], molar ratio [32], deposition pressure [33], substrate temperature [34, 35] etc on the properties of ZnO thin films has been studied in the literature. In this regard, Altuntepe *et al* deposited AZO films with various film thicknesses. They found that the 300 nm AZO film showed enhanced properties such as the highest transmission value, low sheet resistance and good crystalline quality [29]. Mahdhi *et al* deposited GZO films with various (1% to 5%) Ga dopant concentration. They reported that the 3% Ga-doped ZnO film presented a the larger crystalline size, a smoother surface morphology and the lowest film resistivity [30]. The effect of the post annealing approach (RT, 300, 400, and 500 °C) was investigated by Yen *et al* They reported that the post annealing temperature of 300 °C enhanced the properties as the lowest resistivity of 1.36×10^{-4} Ω.cm, optical band gap around 3.82 eV, carrier concentration of 3.95×10^{21} cm^{-3}, the average optical transmittance of 88% at visible range [31]. In addition, Yao *et al* investigated the molar ratio (Ga/Zn molar ratios of 0, 6, 8 and 10%) effect on the GZO thin films. They suggested that 10% Ga/Zn ratio increased the properties like larger resistance ratio and higher device yield [32]. Among these parameters, the deposition pressure and substrate temperature to obtain the films have crucial role on the characteristics of them. The optimization of these parameters can directly affect the crystalline quality of GZO film, which contributes to the formation of larger grain size and less defects [34]. The larger grain size, less

defect, high transparency, and low resistivity can enhance the conversion efficiency of the solar cells [36]. In one study, Park *et al* prepared 200 nm thick GZO layers on quartz using 100, 200, 300, 400 and 500 °C substrate temperatures (T_s) by the PLD method. They reported that the crystalline quality of the films improved, and the band gap shifted from 3.38 to 3.51 eV due to the increase in T_s. According to this study, 300 °C T_s showed the optimum opto-electronic properties [34]. In another study, Wu at al. investigated the impact of T_s on structural and physical properties of GZO films produced by RF magnetron sputtering. They found out that thermal stress decreased by increasing T_s because of the thermal expansion coefficients difference of the film and the substrate. In addition, the crystallite size, resistivity and transmittance healed with the substrate temperature ranging from RT to 300 °C [37]. In their study, Cuong and coworkers fabricated GZO thin films at different deposition pressures via magnetic sputtering system. According to this study, carrier concentration and mobility increased while resistivity decreased with the decrease in the deposition pressure [38]. In yet another study, J K Kim *et al* prepared GZO thin film with different thickness using various deposition pressures (10 mTorr to 40 mTorr) and different RF-power density using magnetron sputtering. They revealed that the resistivity of the GZO films decreased whereas the average transmittance value increased over 85% by decreasing the deposition pressure to 15 mTorr [17]. Other scholars, Shin and his coworkers, studied the effect of deposition pressure, substrate temperature and RF power on structural and electrical properties of the polycrystalline GZO thin films by RF magnetron sputtering. They changed the deposition pressure from 3.5 to 7.5 mTorr, the substrate temperature from 250 °C to 400 °C and RF power from 100 W to 200 W. They observed that the lowest resistivity value was obtained using 350 °C, 6 mTorr and 200 W deposition parameters [35]. Depositing GZO thin films on the c-Si solar cell with thickness of 400 nm as a TCE layer, Untila *et al* noted that 400 nm GZO thin film showed the optimum opto-electronic properties for the solar cell structure [39]. Ma *et al* deposited 700 nm thick GZO thin films by the DC magnetron sputtering method. According to their characterization methods, the crystalline size, film resistivity, and optical band gap values enhanced under 1.0 Pa deposition pressure as 27.5 nm, 4.48×10^{-4} Ω.cm, and 3.3 eV, respectively [40]. Appani *et al* investigated the strain behavior of GZO thin films with different oxygen pressures and they revealed that the crystalline size decreased, and tensile strain increased with the increase of O_2 percentage [41].

Considering the reported studies in the literature, it can be noted that deposition pressure and substrate temperature have been examined in detail regarding their effect on the film properties. However, studies investigating both high deposition pressures and low substrate temperatures are very rare. In fact, there is only one such study but unlike this research, in our study we used higher deposition pressure (from 5 mTorr to 20 mTorr) and lower substrate temperatures (from RT to 250 °C) to investigate the impact of high deposition pressures and low substrate temperatures simultaneously. The reason why we preferred a lower temperature is because of the adverse effect it has on solar cell efficiency. Furthermore, the augmentation of film properties in GZO thin films has been correlated with the thicknesses surpassing 400 nm in the given literature above. At the same time, it should be stressed that an elevation in film thickness results in a reduction in film permeability and necessitates the utilization of excessive material. Consequently, meticulous care was exercised in this research to maintain the film thickness under the parameters of the prevailing literature. Finally, we examined both the working pressure and substrate temperature effects on the properties of GZO films and used the most promising GZO films as a TCE film in silicon-based solar cells.

2. Experimental

2.1. Substrate cleaning and film deposition

GZO samples were fabricated using the RF sputtering employing various deposition pressures and substrate temperatures. Glass substrates were used to grow GZO films. All substrate cleaning and deposition parameters are given in table 1. The thickness of GZO films was selected as 350 nm according to our previous optimization reported recently, which includes the thickness calibration of GZO films [3]. Ahead of the deposition of the films, all the substrates were cleaned using standard cleaning parameters (isopropanol, acetone and DI water) by ultrasonic cleaning system. Then, the cleaned substrates were placed into the PVD system. All the samples were fabricated using a single GZO target (95% ZnO + 5% Ga_2O_3) with 4 N purity. The vacuum level before the deposition process was 1.6×10^{-6} Torr and the RF power was adjusted to 50 W.

The deposition pressure of the GZO thin film was changed from 5 mTorr to 20 mTorr with 5 mTorr increment (5, 10, 15, 20 mTorr) at room temperature and they were labeled as GZO-5, GZO-10, GZO-15, and GZO-20 with respect to the deposition pressure (see table 2). Upon the characterization of GZO films prepared under different pressures, GZO-15 was determined to be the most promising sample. Then, the substrate temperature of GZO films was investigated using RT, 150 °C, 200 °C and 250 °C. The samples were then labeled as GZO-15, GZO-15-150, GZO-15-200 and GZO-15-250 with respect to substrate temperatures (see table 1).

Table 1. Cleaning and deposition parameters of GZO thin films.

Cleaning parameters	
Isopropanol	10 min
Acetone	10 min
DI Water	20 min
Deposition parameters	
Base Pressure (Torr)	1.6×10^{-6}
Growth Pressure (mTorr)	5, 10, 15, 20
RF Power (W)	50
Deposition Rate (Å s^{-1})	0.9–1.0
Thickness	350 nm
Substrate Temperature (°C)	RT, 150, 200, 250

Table 2. Abbreviations of GZO thin films with different deposition pressures and substrate temperatures.

Sample ID	Deposition pressure	Substrate temperature
GZO-5	5 mTorr	RT
GZO-10	10 mTorr	RT
GZO-15	15 mTorr	RT
GZO-20	20 mTorr	RT
GZO-15–150	15 mTorr	150 °C
GZO-15–200	15 mTorr	200 °C
GZO-15–250	15 mTorr	250 °C

2.2. Characterization of the films

X-ray Diffraction (XRD, Panalytical - Empyrean) technique was employed to analyze the crystal structure of GZO samples. The surface microstructure of the GZO thin films were evaluated by Atomic Force Microscopy (AFM, Bruker - Innova). The optical transmission measurements were performed using spectroscopic Ellipsometer (J A Woollam - V Vase). As for the electrical characterization of the GZO samples, they were analyzed by Hall measurement system. The most promising GZO thin film was finally determined based on the structural and opto-electrical characterizations.

2.3. Solar cell fabrication and characterization

After performing all the characterizations, GZO-15 and GZO-15-200 samples were chosen for silicon-based solar cells thanks to their outstanding features after the optimization of growth conditions. Afterwards, Si-based solar cell p-n structure was constructed employing plasma enhanced chemical vapor deposition (PECVD) technique by depositing n-type layer on p-type wafer. Subsequently, the GZO-15 and GZO-15-200 films were deposited with PVD system onto the Si-based p-n structure as TCE layer. Finally, the metallization were implemented utilizing silver paste by screen printing method to finalize the cell structure. The schematic structure of the cell is illustrated in figure 1. The solar simulator with AM-1.5 and 100 mW cm^{-2} was used to characterize the solar cells (2.5 cm × 2.5 cm).

3. Results and discussion

3.1. The effect of deposition pressures

In the present study, the effect of deposition pressure (5 mTorr–20 mTorr) on the crystal structure of GZO samples was examined by analyzing XRD patterns.

The XRD pattern of GZO samples prepared under various deposition pressures is given in figure 2. As presented in this figure, the XRD patterns of GZO layers were predominated by the peaks placed at around $2\theta = 33.82°$ and 62.70° corresponding the (002) and (103) diffraction planes of ZnO-based hexagonal wurtzite structure, respectively (JCPDS 036-1451) [42]. As the deposition pressure increased, the diffraction peaks shifted towards the higher angle (5 mTorr: 33.82° to 10 mTorr: 34.02°, 15 mTorr: 34.05°, 20 mTorr: 34.03°) (see figure 2). The observed peak shifting toward higher angles corresponds to in-plane compressive stress in the film, which decreased the interplanar distance, d [43]. There is an inverse proportion between interplanar distance and shifted peak positions. Adding smaller ions in the host lattice leads to an decrease in the interplanar distance,

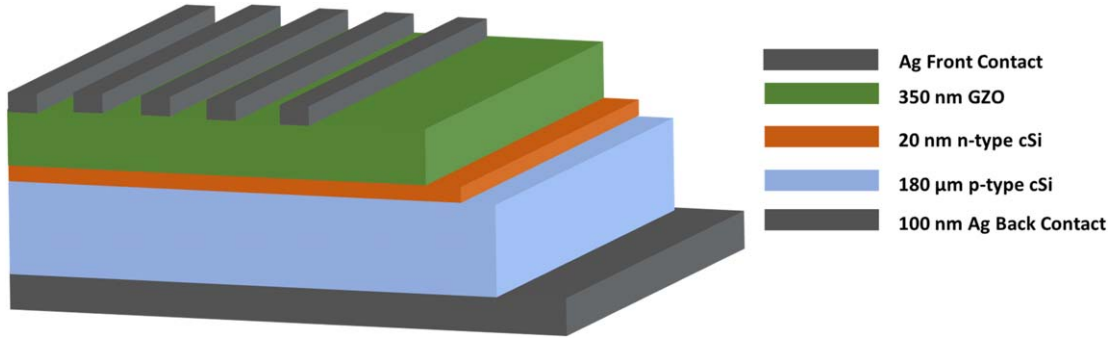


Figure 1. Schematic structure of Si-based solar cell constructed by utilizing GZO thin film.

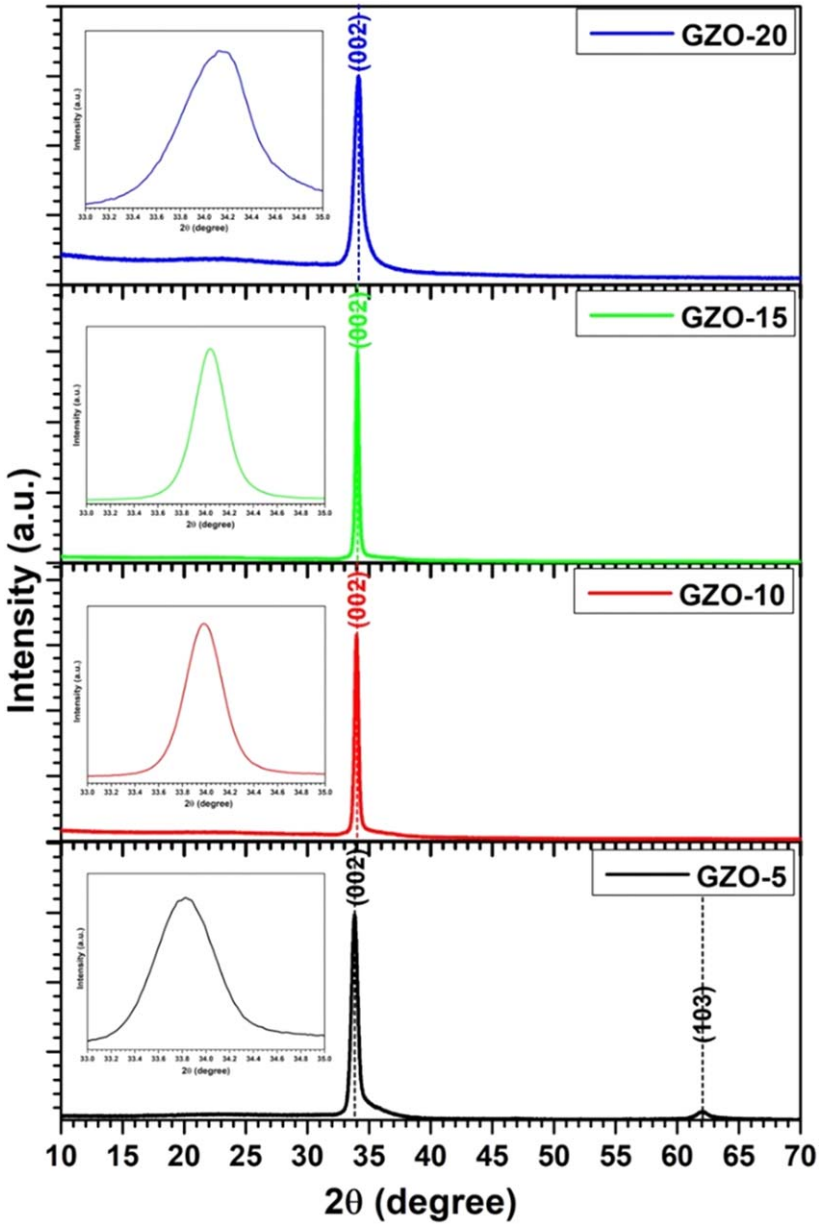


Figure 2. XRD pattern of GZO samples deposited under distinct pressures and zoomed in (002) diffraction peak (inset).

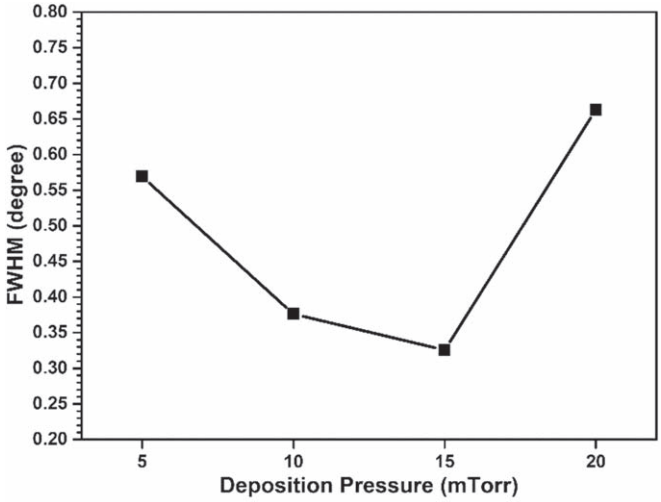


Figure 3. The FWHM values of GZO samples fabricated under various deposition pressures.

which could be assigned to the substitution of Zn^{2+} by Ga^{3+} due to the difference in atomic radius of Zn^{2+} (0.75 Å) and Ga^{3+} (0.62 Å) [44].

The FWHM values of (002) preferential diffraction peak received from figure 2 are presented in figure 3. The purpose here was to investigate the influence of working pressure on the crystallite size of GZO samples**.** It is well known that the FWHM values are strongly associated with the crystalline quality of thin films. Therefore, the crystallite size was determined by using Scherer's equation [45].

$$D = \frac{(0.9)^{*}\lambda}{\beta \cos\theta} \tag{1}$$

where D, λ, β and θ represents the crystal size, wavelength of CuK_{α} radiation (1.54056 Å), the FWHM of the (002) peak and the Bragg angle, respectively. According to equation (1), the crystal quality is inversely proportional with the FWHM values, which means the lower FWHM value corresponds to the larger crystallite size. As seen in figure 3, the FWHM value decreases with increasing the pressure up to 15 mTorr and then it starts to increase. This trend demonstrated that the crystallite size of GZO films was enhanced with the rise in deposition pressure up to 15 mTorr. However, it started to worsen at higher pressure. According to the calculation, the crystal size increased from 15.25 nm to 26.65 nm as the deposition pressure elevated from 5 mTorr to 15 mTorr. Then, it decreased to 13.10 nm for 20 mTorr deposition pressure. Considering the XRD data, it can be summarized that the GZO thin film fabricated under 15 mTorr deposition pressure revealed more promising crystalline quality. Deterioration of the crystal size with increasing the growth pressure from 15 to 20 mTorr can be associated with the increase striking Ar ions with deposited atoms [46, 47].

The interplanar distance (d_{hkl}) of GZO thin films was calculated from Bragg equation (equation (2)). Bragg equation is provided below;

$$n\lambda = 2d \sin(\theta) \tag{2}$$

where, n is the order of diffraction, λ is the wavelength of the incident X-rays, d is the interplanar spacing, and θ is the Bragg angle. As the deposition pressure changed, so did the interplanar distance. The interplanar distance of GZO thin films was found as 2.6517 Å for GZO-5, 2.6351 Å for GZO-10, 2.6217 Å for GZO-15 and 2.6341 Å for GZO-20. The interplanar distance decrease could be ascribed to the diffusion of Ga atoms to the ZnO lattice by increasing the deposition pressure up to 15 mTorr and then interplanar distance slightly increased due to increase in the collision of atoms. In other words, 15 mTorr seems the optimum deposition pressure for the Ga atoms diffusion into the host. Besides, we observed that the (103) diffraction peak was seen only in the XRD pattern of GZO thin film prepared under 5 mTorr working pressure. Increase in deposition pressure above 5 mTorr brought about to the disappearance of (103) the diffraction peak and contributed to the crystallization of GZO films only through (002) plane direction. It means that increasing the deposition pressure may change the crystal orientation of GZO films.

The micro-strain and dislocation density values of all samples calculated by using equation (3) [48] and (4) [49] as below,

$$\varepsilon = \frac{\beta \cos\theta}{4} \tag{3}$$

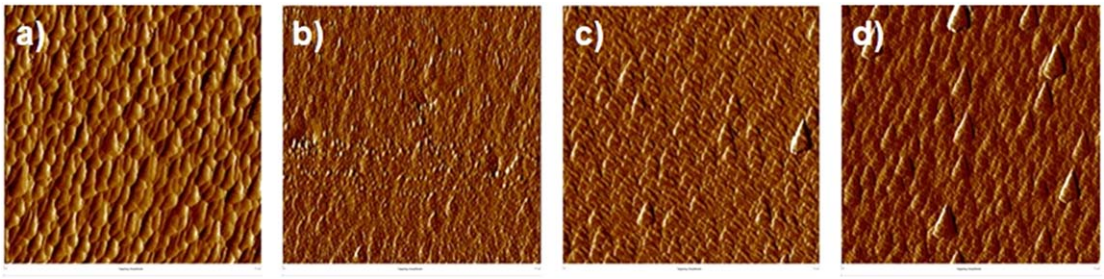


Figure 4. The AFM measurement of the GZO films prepared with different deposition pressures (a) GZO-5, (b) GZO-10, (c) GZO-15 and (d) GZO-20.

Table 3. The strain and dislocation density of pressure dependent GZO thin films.

Sample ID	ε	δ (nm^{-2})
GZO-5	2.38×10^{-3}	4.3×10^{-3}
GZO-10	1.54×10^{-3}	1.85×10^{-3}
GZO-15	1.33×10^{-3}	1.41×10^{-3}
GZO-20	2.79×10^{-3}	5.83×10^{-3}

$$\delta = \frac{1}{D^2} \tag{4}$$

where D, β, ε, δ, and θ represents the crystal size, the FWHM of the (002) peak, micro-strain, dislocation density and the Bragg angle, respectively. Calculated ε and δ of pressure dependent films are given in table 3. The ε of the pressure dependent GZO films decreased from 2.38×10^{-3} to 1.33×10^{-3} upon the increase in the deposition pressure from 5 mTorr to 15 mTorr. Then, it increased to 2.79×10^{-3} for 20 mTorr. The increase or decrease of ε can induce a rearrangement of atomic positions and increase the (002) preferable orientation in the lattice, thereby altering all the physical properties of the films. In the literature, the shifting of (002) peak positions to the higher angles triggered a decrease in the ε and δ of the GZO thin films [50]. Moreover, the δ of pressure dependent GZO films decreased from 4.3×10^{-3} to 1.41×10^{-3} nm^{-2} with the increased deposition pressure from 5 mTorr to 15 mTorr. The δ is associated with the defect density in the structure and the lower defect caused the lower δ [49].

Figure 4 represents the AFM surface measurement of the GZO films with the scan size of 5 μm $\times$ 5 μm. The AFM measurements showed that all the samples had a dense surface morphology regardless of the deposition pressure. However, the surface of the GZO-5 thin film displayed inhomogeneous microstructure distribution. As the growth pressure increased (5 mTorr to 15 mTorr), the film surface displayed a more homogenous, compact, and denser structure. As well as the surface morphology, the surface roughness (R_{rms}) was also evaluated through the AFM measurement. The R_{rms} values were found out to be 4.8 nm, 6.2 nm, 8.3 nm, and 3.8 nm for GZO-5, GZO-10, GZO-15 and GZO-20 samples, respectively. In the literature, increasing the R_{rms} value was attributed to the enhancement of the crystallite size [51]. This result supports the XRD data of GZO samples growth at different pressures.

The transmission measurement results of GZO thin films is provided in figure 5 and the average values in the visible region (400–800 nm) of the electromagnetic spectrum are given in table 4. When the deposition pressure of GZO thin films increased from 5 mTorr to 15 mTorr, the transmission values of the sample did not significantly alter in the electromagnetic spectrum (~82%). However, the rise in the growth pressure from 15 mTorr to 20 mTorr gave rise to a decrease in the transmission from 82% to 80%. This slight decrease in optical transmission can be ascribed to the deterioration in crystal quality. Table 4 shows the highest transmission values of GZO films in the electromagnetic spectrum.

Moreover, the absorption coefficient (α) of GZO thin films was calculated by using Lambert–Beer law [52]. The relation is described in equation (5),

$$\alpha = \frac{1}{d} \ln\left(\frac{1}{T}\right) \tag{5}$$

where d is the thickness and T is the transmission data of the films. After calculating the α, the Tauc plot [53] was employed to estimate energy band gap of the samples using equation (6).

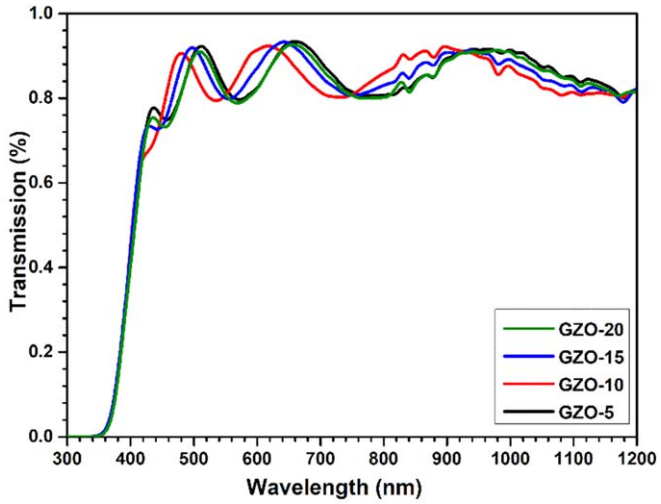


Figure 5. The transmission data of GZO samples grown under distinct growth pressures.

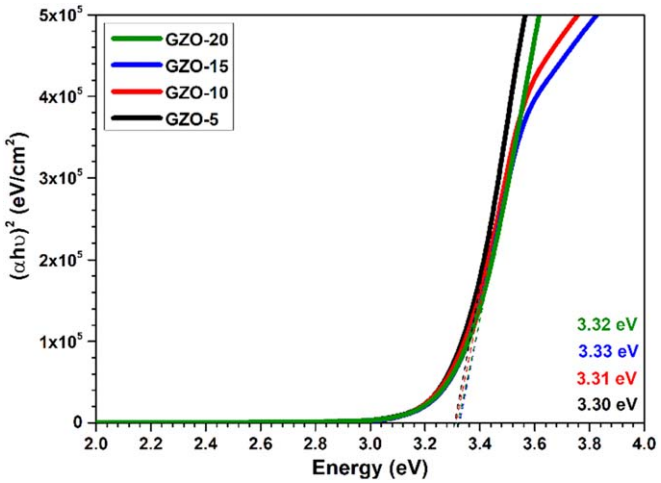


Figure 6. Estimation of band gap of GZO samples by $(\alpha h\upsilon)^2$ versus $h\upsilon$ photon energy (eV) plot.

Table 4. Average optical transmission (400–800 nm) of GZO thin films.

Sample	Transmission (%)
GZO-5	82.4
GZO-10	82.2
GZO-15	82.6
GZO-20	80.3

$$(\alpha h\upsilon) = A(h\upsilon - E_g)^{1/2} \quad (6)$$

where A is the constant, $h\upsilon$ is the photon energy, E_g is the optical band gap energy presented. The optical band gap was determined by $(\alpha h\upsilon)^2$ versus $h\upsilon$ photon energy (eV) plot, which is presented in figure 6. The band gap value of the GZO-5 film was determined as 3.30 eV. After the deposition pressure increased up to 15 mTorr, the band gap value increased to 3.33 eV. The increase of the pressure above 15 mTorr gave rise to a decrease to 3.32 eV for GZO-20 film. The broadening E_g can be attributed to the Burstein–Moss Effect [54], where energy states closed to conduction band may be occupied by free electron concentration and this leads to energy band shifting [55]. This higher concentration of free electrons can be associated with the presence of a higher amount of Zn_i defects, specifically shallow donors, within the GZO film structure. The presence of these Zn_i defects leads

Table 5. Some electrical parameters of GZO samples prepared under distinct deposition pressures.

Sample	GZO-5	GZO-10	GZO-15	GZO-20
Conductivity	n	n	n	n
Resistivity (Ω.cm)	1.03	4.24×10^{-2}	5.33×10^{-2}	5.46×10^{-2}
Carrier Concentration (cm^{-3})	1.02×10^{18}	6.39×10^{19}	7.22×10^{19}	4.76×10^{19}

to additional energy states within the bandgap of the material, which can capture and release electrons. These shallow donor states are typically associated with the presence of excess zinc or gallium ions or oxygen vacancies in the crystal lattice of GZO films. The higher deposition pressure (15 mTorr@RT) during film growth can influence the formation of these defects, resulting in an increased concentration of shallow donors. The shallow donors lead to a larger free electron concentration in the GZO films and are likely to be responsible for the observed band shifting in the pressure dependence measurements [56]. The higher band gap in TCE films allows more photons to reach the solar cell structure, and thus improves the conversion efficiency [57]. Therefore, the GZO thin film prepared under 15 mTorr presented a relatively more suitable and efficient band gap for potential solar cell applications (GZO-15).

The electrical properties (resistivity, carrier concentration etc) of TCE materials are critical because charge carriers are collected over this layer. The conduction type, resistivity (Ω.cm), and carrier concentration (cm^{-3}) values of GZO samples are summarized in table 5. Regardless of the deposition pressure, all the films showed n-type conductivity. We observed that the rise in the growth pressure above 5 mTorr caused a considerable decrease in the resistivity of the films. The resistivity changed from 1.03 Ω.cm to 4.24×10^{-2} Ω.cm with the rise in the growth pressure. The deposition pressure above 5 mTorr in GZO films showed similar resistivity values that is on the level of 10^{-2} Ω.cm. Similarly, the carrier concentration values enhanced with the rise in the growth pressure above 5 mTorr. The density of charge carriers increased from 1.02×10^{18} to 7.22×10^{19} cm^{-3} upon rising the deposition pressure. Although the carrier concentration values of GZO-10, GZO-15 and GZO-20 are very close, the highest value was obtained with GZO-15 thin film. The highest charge carrier density of GZO-15 sample can be ascribed to increasing diffusion of Ga atoms through the Zn sites from interstitial locations and grain boundaries [58].

Overall, as a result of the analyses of GZO samples produced under different deposition pressures, the GZO-15 sample was determined to be the most promising TCE film in terms of enhanced crystalline quality, higher transmission and optical band gap, and better electrical properties. Therefore, in the last part of this research, the effect of substrate temperature was examined using GZO-15 sample. To do so, various temperatures (RT, 150 °C, 200 °C and 250 °C) were used. For example, while the GZO-15-RT sample represents the sample prepared using 15 mTorr and room temperature, the GZO-15-250 sample represents the one prepared using 15 mTorr and 250 °C substrate temperature. It should be noted that the GZO-15 and GZO-15-RT samples represent the same sample since both were produced under the same conditions.

3.2. The effect of substrate temperature

The impact of substrate temperature (RT, 150 °C, 200 °C and 250 °C) on the crystal structure of GZO-15 thin films was evaluated using XRD patterns. The XRD patterns of GZO films are illustrated in figure 7. As observed in figure 7, the diffraction patterns were governed by the (002) diffraction peaks corresponding to hexagonal wurtzite ZnO structure (JCPDS 036-1451) [42]. These diffraction peaks slightly shifted towards higher angles by increasing temperature from RT to 200 °C. These shifts are attributed to the interplanar distance as observed in GZO films prepared under distinct deposition pressures. When substrate temperature was risen from RT to 200 °C, the interplanar distance decreased from 2.6217 Å to 2.6138 Å. Increasing the temperature above 200 °C contributed to an increase in the interplanar distance up to 2.6308 Å. The shift of the interplanar distance can be ascribed to variation of Ga diffusion into host ZnO structure as a result of the substrate temperature owing to ionic radius difference between Zn^{2+} and Ga^{3+} [59].

The variation of FWHM value of (002) diffraction peaks of GZO films fabricated under different temperatures is illustrated in figure 8. Elevating the temperature from RT to 200 °C ended up with a decrease in the FWHM of GZO thin films, which we attribute to the enhancement in the crystallite size which was calculated as 26.65 nm, 34.15 nm, 51.71 nm, and 49.63 nm for GZO films deposited at RT, 150 °C, 200 °C and 250 °C, respectively. This result showed that the best crystallite size can be obtained in GZO films deposited at 200 °C.

Calculated ε and δ of temperature dependent GZO films are given in table 6. The ε of the temperature dependent GZO films decreased from 1.33×10^{-3} to 7.09×10^{-4} when the substrate temperature was risen from RT to 200 °C. Moreover, the δ of temperature dependence GZO films decreased from 1.41×10^{-3} to 3.74×10^{-4} nm^{-2} with the increased substrate temperature from RT to 200 °C. Then, it increased to

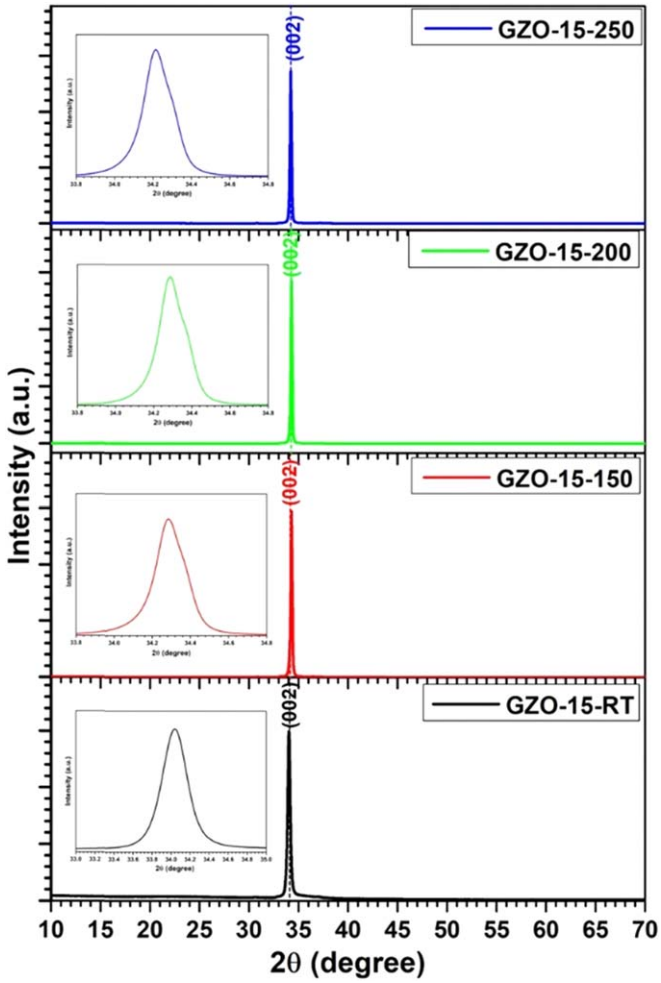


Figure 7. XRD pattern of GZO thin samples grown at various substrate temperatures and zoomed in (002) diffraction peak (inset).

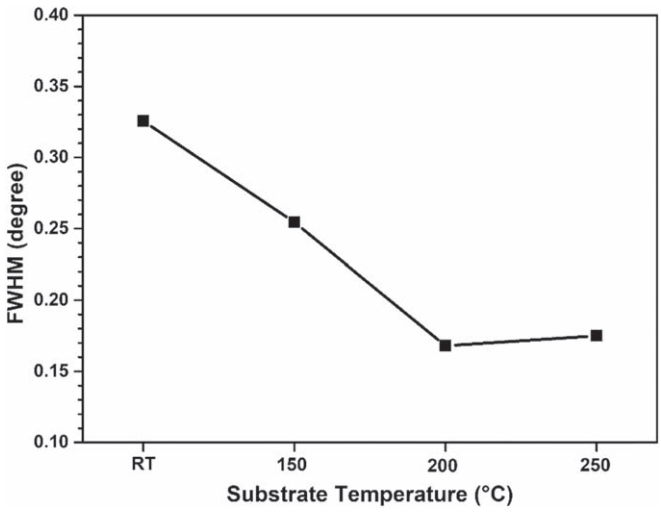


Figure 8. The FWHM value of the GZO samples prepared at different temperatures.

4.06×10^{-4} for 250 °C. According to the D, ε, and δ values of the all samples, the GZO-5-200 sample showed the best structural properties.

Figure 9 presents the AFM surface morphology (5 μm × 5 μm) of GZO thin films prepared at various temperatures. The evolution of surface morphologies clearly changed upon increasing substrate temperature.

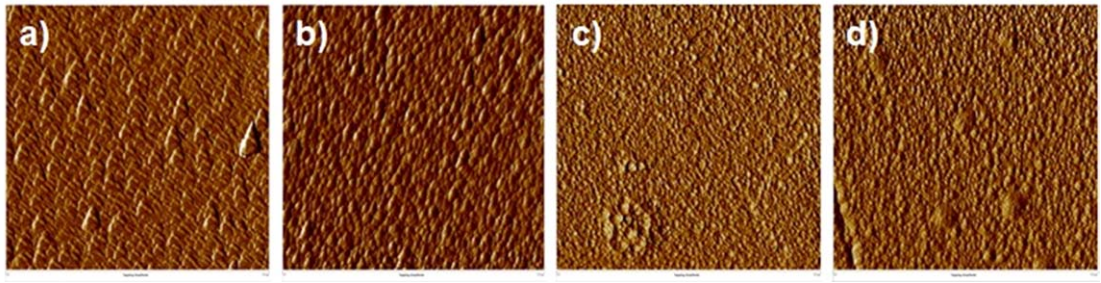


Figure 9. The AFM measurement of the GZO samples grown at different substrate temperatures; (a) RT, (b) 150 °C, (c) 200 °C and (d) 250 °C.

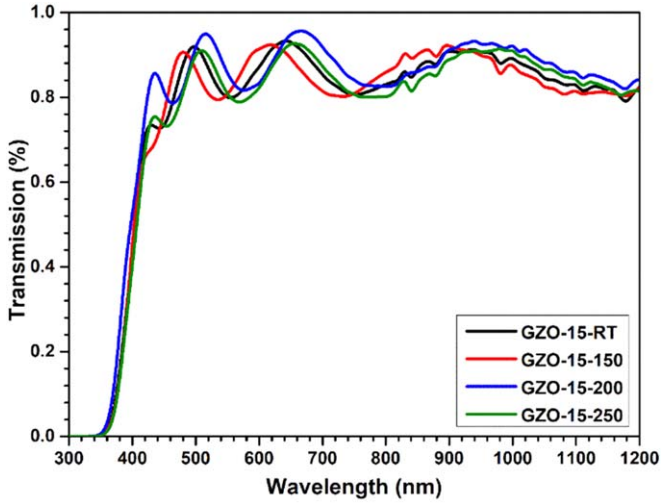


Figure 10. Optical Transmission data of GZO samples grown at different substrate temperatures.

Table 6. The strain and dislocation density of temperature dependent GZO thin films.

Sample ID	ε	δ (nm^{-2})
GZO-5-RT	1.33×10^{-3}	1.41×10^{-3}
GZO-5-150	1.08×10^{-3}	8.57×10^{-4}
GZO-5-200	7.09×10^{-4}	3.74×10^{-4}
GZO-5-250	7.92×10^{-4}	4.06×10^{-4}

When temperature rose up to 200 °C, the surface of films exhibited a denser microstructure with high homogeneity. The R_{rms} values were found as 8.3 nm, 13.4 nm, 18.6 nm, and 16.7 nm for GZO films grown at substrate temperatures of RT, 150 °C, 200 °C and 250 °C, respectively. Wang *et al* reported that the crystal size change is directly proportional to the R_{rms} value that complies with the XRD data of the samples [51].

The transmission measurements (from 300 nm to 1200 nm) of GZO thin films are provided in figure 10 and the average transmission values ranging from 400 nm to 800 nm, which is visible region, are presented in table 7. Elevating the temperature up to 200 °C contributes to enhancing the optical transmission from 82% to 86% and then decreased to 82%. This variation is correlated with the crystal quality of the samples [60]. The highest optical transmission was obtained by GZO-15–200 thin film with the best crystalline quality.

The optical band gap values of GZO thin films were calculated using transmission data and they are presented in figure 11. We can observe that the band gap values fluctuated between 3.32 eV and 3.36 eV, which can be explained through the Burstein–Moss Effect [61]. As seen at Hall and band gap measurements, the higher substrate temperatures (200 °C) can influence the formation of Zn_i defects, resulting in an increased concentration of shallow donors. This larger free electron concentration in the temperature dependence GZO films is responsible for the observed band shifting [62]. The increase in the E_g can also be ascribed to

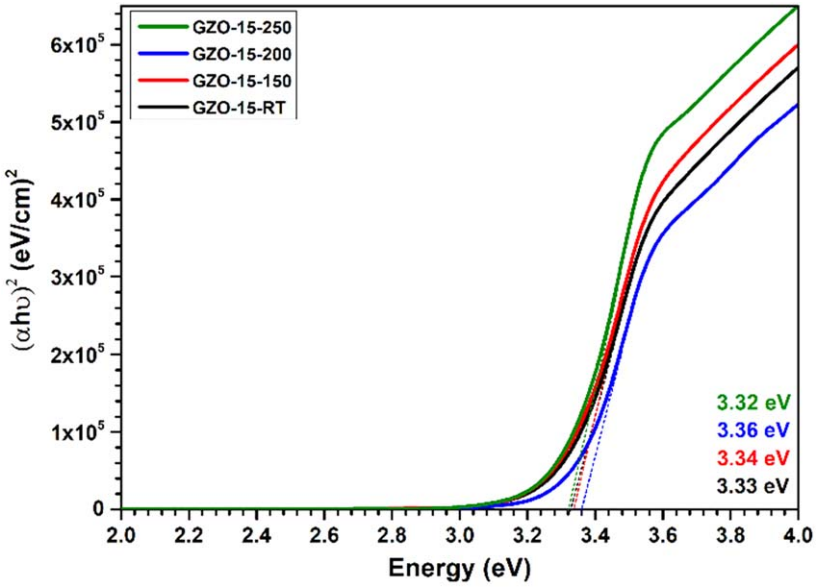


Figure 11. Band gap estimation of GZO thin films grown at various substrate temperature.

Table 7. Average optical transmission (400–800 nm) data of GZO thin films deposited at distinct substrate temperatures.

Sample	Transmission (400–800)
GZO-15-RT	82.6
GZO-15–150	82.2
GZO-15–200	86.8
GZO-15–250	85.4

Table 8. Some electrical parameters of GZO samples grown at different substrate temperatures.

Sample	GZO-15-RT	GZO-15–150	GZO-15–200	GZO-15–250
Conductivity	n	n	n	n
Resistivity (Ω.cm)	5.3×10^{-2}	3.7×10^{-3}	2.0×10^{-3}	3.0×10^{-3}
Carrier Concentration (cm^{-3})	7.2×10^{19}	1.2×10^{20}	1.6×10^{20}	1.5×10^{20}

enhancement in crystalline quality that is related to the lower defective states in which are located below the conduction band. The highest band gap value was obtained by GZO-15–200 thin film with 3.36 eV. This result revealed that such a film is more proper for solar cell applications as a TCE layer due to empowering the transmission of more photons. As reported by Benramache *et al* the band gap shifting is strongly related to the enhancement in the crystal size and increase in doping concentration of ZnO thin films [63].

The electrical properties of GZO thin films are presented in table 8. All the films showed n-type conductivity. However, increasing the temperature brought about decreasing the resistivity from 5.3×10^{-2} to 2.0×10^{-3} Ω. cm, and increasing the carrier concentration from 7.2×10^{19} cm^{-3} to 1.6×10^{20} cm^{-3}. The highest carrier concentration and lowest resistivity values were obtained by GZO film prepared at 200 °C substrate temperature (GZO-15-200), which is associated with the enhanced crystalline quality. Interstitial atoms and scattering occurred in the grain boundaries may decrease as the temperature is increased up to 200 °C [64].

3.3. Solar cells

Amongst the GZO samples grown at different deposition pressures and substrate temperatures, the one deposited with 15 mTorr and 200 °C showed the most promising structural, optic and electric properties. On the other hand, in order to evaluate and see the improvement of the cell conversion efficiency with respect to substrate temperature we used the best two TCE samples namely GZO-15 and GZO-15-200, which had been

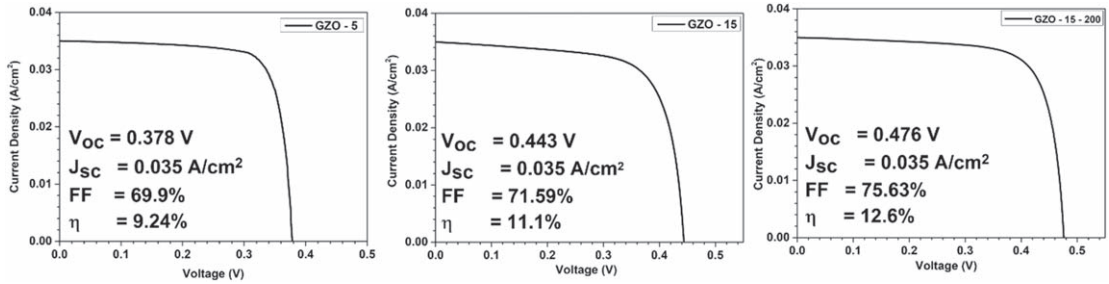


Figure 12. The J-V graphs of solar cells utilizing GZO-5, GZO-15 and GZO-15–200 as TCE layers.

selected based on the growth pressure and substrate temperature optimizations for silicon-based solar cell applications. Additionally, in order to confirm the effect of the pressure on the cell efficiency, a cell was also fabricated using GZO deposited at 5 mTorr@RT. The reason we chose this sample for comparison is improvement of the crystalline quality and opto-electronic properties along with increasing deposition pressure from 5 to 15 mTorr. The cell fabricated with this layer was employed as reference to evaluate the impact of growth pressure on the cell efficiency. The current density (J) versus voltage (V) plots of those cells are illustrated in figure 12. From these plots, we determined some solar cell parameters. The J-V curve of the solar cell by using GZO-5 layer presented 0.378 V, 0.035 mA cm^{-2}, and 9.24% of open-circuit voltage (V_{oc}), short-circuit current density (J_{sc}) and conversion efficiency (η), respectively. The cell with GZO-15 TCE layer exhibited 0.443 V, 0.035 mA cm^{-2}, and 11.1% of V_{oc}, J_{sc}, and conversion efficiency, respectively. On the other hand, the J-V curve of the solar cell constructed by using GZO-15-200 TCE layer demonstrated 0.476 V, 0.035 mA cm^{-2}, and 12.6% of V_{oc}, J_{sc}, and cell efficiency, respectively. The results clearly showed that the increase in deposition pressure and temperature led to increase in the cell efficiencies. Depending on the comparison of the solar cells, we can deduce that the V_{oc} (from 0.378 V to 0.476 V), FF (from 69.9% to 75.63%) and conversion efficiency (η) (from 9.24% to 12.6%) enhanced upon applying higher pressure and higher substrate temperature treatment to GZO TCE layer. The increase in the V_{oc} can be associated with the crystallite size and opto-electrical properties enhancement of GZO thin film. Thus, it can be related to the decrease in the recombination of charge carriers thanks to enhancing the grain size along with decreasing the grain boundaries which behave as recombination/trap centers for the charge carriers [65]. On the other hand, since the J_{sc} in silicon heterojunction solar cells independent of TCE, J_{sc} stays constant (only minor differences) for all the studied cells, as expected [66]. When the deposition pressure was increased to 15 mTorr (@RT) and the substrate temperature to 200 °C (@15mTorr), the film resistivity reduced while the carrier concentration enhanced. This lower film resistivity and enhanced carrier concentration caused an increase in the FF value [67].

4. Conclusion

In this study, the effect of deposition pressure and substrate temperature on the structural, optical and electrical properties of Ga doped ZnO samples and the photovoltaic performance of this structure as a TCE layer in silicon-based solar cell were examined. All the GZO samples were grown by the RF magnetron sputtering approach using GZO target. These films were analyzed by employing several techniques such as XRD, AFM, Optical Transmission Spectroscopy, Hall measurement, Sun simulator. The data obtained from these measurements were used to calculate some film parameters like band gap, interplanar spacing, crystalline size, micro-strain, dislocation density. All the prepared films was dominated by (002) preferential orientation regardless of the deposition pressures and substrate temperatures. Increasing the growth pressure and substrate temperature contributes to improving the crystallite size, widening the optical band gap, enhancing the electrical parameters. In addition, the micro-strain and dislocation density of films decreased with the increased deposition pressures from 5 mTorr to 15 mTorr. Following this, all the properties of the films were enhanced thanks to the increased substrate temperature from RT to 200 °C. Among the film grown with different pressures, GZO-15 was chosen as the promising film with 26.65 nm crystalline size, 1.34×10^{-3} micro-strain, 1.41×10^{-3} nm^{-2} dislocation density, 5.33×10^{-2} Ω.cm resistivity, and 7.22×10^{19} cm^{-3} carrier concentration values. Later, the samples were grown at 15 mTorr with different substrate temperature (RT, 150, 200, and 250 °C). GZO-15–200 sample showed enhanced properties as 51.71 nm crystalline size, 7.09×10^{-4} micro-strain, 3.74×10^{-4} nm^{-2} dislocation density, 2.0×10^{-3} Ω.cm resistivity and 1.6×10^{20} cm^{-3} carrier concentration values. When all the performed analyses are considered, the GZO films obtained with 15 mTorr at RT (GZO-15) and with 15 mTorr at 200 °C (GZO-15-200) were determined as the most promising TCE layer for

solar cell applications from the growth optimizations. The photovoltaic performance of silicon-based solar cell with employing GZO thin film as a TCE layer showed that applying both pressure and substrate temperature to GZO film contributed to enhancing the solar parameters (V_{oc} and FF). Thus, the conversion efficiency increased from 9.24% to 12.6%. Overall, based on our study outcomes, it can be concluded that the deposition pressure and substrate temperature play a significant role on the properties of GZO thin films and its photovoltaic performance of solar cell. Moreover, the GZO film can be used as a promising alternative TCE layer in thin film solar cells replacing vulnerable and costly ITO. Although Ga is a rare material in the earth crust, it is more abundant than In. Furthermore, by optimizing the growth parameters which concern the pressure and temperature of GZO thin films, we can conclude that the thinner film can be fabricated by consuming less materials. Besides that, they have a lower environmental impact. They, therefore, were employed as TCE layer in the solar cell with the cell efficiency that can be competitive with thicker films.

Data availability statement

The data cannot be made publicly available upon publication because no suitable repository exists for hosting data in this field of study. The data that support the findings of this study are available upon reasonable request from the authors.

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